Anteater Chess Ver 1.0

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Glossary

Board - The 10x8 grid on which Anteater Chess is played, with columns labeled A–J and rows labeled 1–8.

File - A column on the board, labeled A through J.

Rank - A row on the board, numbered 1 through 8.

White/Black - The two sides in Anteater Chess. White always moves first.

Ant (Pawn) - A piece that can move 1-2 spaces straight forward on its first turn and 1 space forward on every turn after. The ant captures diagonally.

Anteater - A special piece unique to Anteater Chess. Moves one square in any direction like a King. When capturing, the Anteater can eat multiple ants (pawns) in a single move, as long as they are adjacent to each other horizontally or vertically (not diagonally)

Ant eating - a special move that allows the anteater piece to continuously capture ants (pawns) who are adjacent to each other in the same file or rank but not diagonally

Bishop - A piece that moves diagonally any number of times.

King - The most important piece on the board. Moves one square in any direction. If the King is checkmated, the game is over.

Knight - A piece that moves in an L-shape: two squares in one direction and one square perpendicular. The only piece that can jump over other pieces.

Queen - The most powerful piece on the board. Can move any number of squares horizontally, vertically, or diagonally.

Rook - A piece that moves any number of squares horizontally or vertically.

Capture - The act of moving a piece onto a square occupied by an opponent's piece, removing it from the board.

Castling - A special move involving the king and rook. It is allowed only if neither piece has moved, the squares between them are empty, the king is not in check, and the king does not move through or end on a square under attack.

Check - A situation where a player's King is under direct threat of capture on the opponent's next move. The player must resolve the check immediately.

Checkmate - A situation where a player's King is in check, and there is no legal move to escape. The game ends, and that player loses.

En Passant - A special pawn capture that can occur immediately after an opponent's pawn moves two squares forward from its starting position. The capturing pawn moves to the square the opponent's pawn passed through.

Legal Move - Any move that follows the rules of Anteater Chess and does not leave the player's own King in check.

Stalemate - A situation where a player has no legal moves available, but their King is not in check. The game ends with no winner.

1. Computer Chess

1.1 Usage Scenario

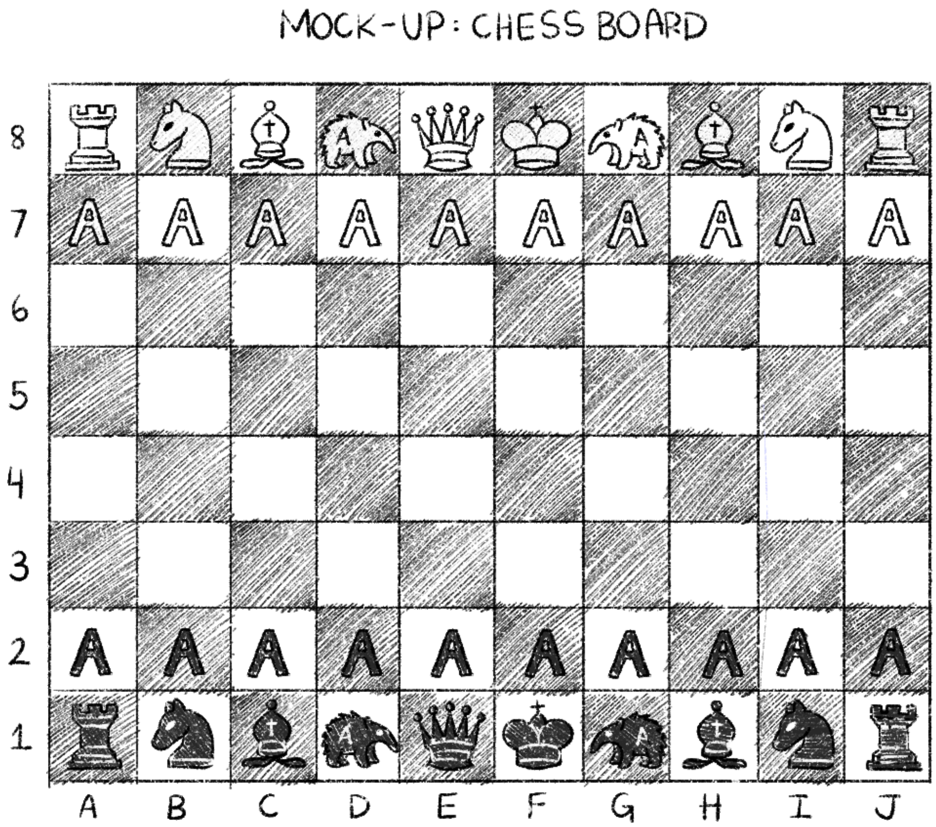

Fig. 1.1: Mock-Up ChessBoard: Sketched

A human user invokes the Anteater Chess program and is taken to the main menu. The user selects "Start New Game" and wants to play as white against the computer. The graphical user interface (GUI) shows the initial position of the board. The user chooses their pawn on F2 and selects F4 as the destination for his or her first move. The computer instantly responds by moving a black pawn. This takes less than 1 minute to calculate. The game continues interactively and logs all moves into a text file until one player's King is in checkmate. At one point in the game, the user moves their Anteater to capture a chain of adjacent enemy ants in a single move, demonstrating the unique mechanics of Anteater Chess.

1.2 Goals

The goal of Anteater chess is to provide the player with a playable computer version of chess with added features on top of the original game. It will allow the player to play an interactive game of Anteater chess against a computer while providing a seamless, user-friendly experience. The program will also enforce the rules of standard Chess and the new ones presented in Anteater Chess. In addition to standard chess rules, the program introduces the Anteater piece, a unique addition that adds a new layer of strategy to the game.

1.3 Features

Current features:

- Realistic, interactive Human vs Computer gameplay.
- Graphical interface of the board and pieces in the current position.
- Chess and Anteater chess rule enforcement.
- Classic chess moves.
- Option to choose player color as the black or white side.
- Readable text file logging moves.
- Responsive, realistic computer opponent.
- Legal move enforcement (illegal moves are rejected with an error message)
- En passant and castling support
- Pawn/ant promotion

Future Potential Features:

- Human vs Human and Computer vs Computer modes.
- Undo feature for players' moves.
- Different difficulties of gameplay: beginner, intermediate, and expert.
- Hints for the correct or highest probability of success move.
- Option for turn time limits.
- The option to save and load games.
- Supports algebraic notation of chess moves.
- Chess clock/timer display.

2. Installation

2.1 System Requirements

Operating System: Linux OS (compatible with servers like bondi, laguna, or crystalcove)
Compiler: Standard C Compiler (gcc) and standard C libraries.
Libraries: Appropriate GUI libraries (will be introduced in Lecture 4).

2.2 Set Up and Configuration

Installing the Program

If you have the binary package, extract it and run the program with these commands:
$ tar -xvf binaryArchive.tar.gz
$ cd chess
$ ./chess
If you'd rather build from source, use these commands instead:
$ tar -xvf sourceArchive.tar.gz
$ cd chess
$ make
$ ./chess

Starting the Game

When you first launch the program, you'll be asked to choose your color (White or Black) and a difficulty level for the AI opponent. White always goes first. Once you've made your selections, the board will appear, and the game will begin.

Board Basics

The Anteater Chess board uses columns A through J and rows 1 through 8. To make a move, type the starting square followed by the destination square - for example, E2 E4. The program will tell you if a move is invalid.

2.3 Uninstalling

To remove Anteater Chess from your system, simply delete the chess/ directory that was created during installation:
$ rm -rf chess
This will remove the program, all its files, and any saved data.

3. Chess Program Functions and Features

3.1 Menu Function

When paused, the menu displays and the user can select to start a new game, resume a game, Quit, etc., and the program will output the corresponding ‘screen.’ For ‘start a new game’, the board will reset. For ‘resume a game’, the menu display disappears and the board will display as previously. For ‘Quit’, the program will display the initial screen (no board).

```
Paused Menu:

  1. Resume
  2. Start New Game
  3. Save Game
  4. Quit

Please input the number of which option you'd like to choose:
```

Fig. 3.1: Paused Menu to be displayed when Menu Function is run

3.2 Board Display Function

Each time the user inputs a new move or the bot makes a move, the program will display an updated board.

```
   +----+----+----+----+----+----+----+----+----+----+
 8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
   +----+----+----+----+----+----+----+----+----+----+
 7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
   +----+----+----+----+----+----+----+----+----+----+
 6 |    |    |    |    |    |    |    |    |    |    |
   +----+----+----+----+----+----+----+----+----+----+
 5 |    |    |    |    |    |    |    |    |    |    |
   +----+----+----+----+----+----+----+----+----+----+
 4 |    |    |    |    |    |    |    |    |    |    |
   +----+----+----+----+----+----+----+----+----+----+
 3 |    |    |    |    |    |    |    |    |    |    |
   +----+----+----+----+----+----+----+----+----+----+
 2 | wP | wP | wP | wP | wP | wP | wP | wP | wP | wP |
   +----+----+----+----+----+----+----+----+----+----+
 1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
   +----+----+----+----+----+----+----+----+----+----+
     A    B    C    D    E    F    G    H    I    J

 [1 to Pause Game, 2 to Undo Previous move, 3 for Hint]
```

Fig. 3.2: Board Updates after Board Display Function

3.3 Piece Move Function

The user chooses a specific piece and chooses the target destination square. It will check if the move is legal and if it is, the piece chosen will be moved to a new location. If the Anteater moves to make an 'ant eating' move, the program captures the target ant and any additional ants that are horizontally or vertically adjacent to each other in a chain from that position. Diagonal adjacency does not count

```
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP | wP | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J
```

[1 to Pause Game, 2 to Undo Previous move, 3 for Hint]

Type the coordinate of which piece to be moved in format of ‘A2’(or 1/2/3): F2

Type the coordinate of where the piece will be moved in format of ‘A3’: F3

```
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    | wP |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP |    | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J
```

[1 to Pause Game, 2 to Undo Previous move, 3 for Hint]

Fig. 3.3: Before & After of Piece Move Function

3.4 Legal Move Check Function

Checks if the user/bot's move is legal, depending on the specific piece. If the move is not legal, the program will display a message informing the user that the move is not possible and to try again. If the move is legal, then the game will continue as normal.

- Can the specific piece move in that direction or that many spaces?
- Is there a piece that is blocking them?

```
Illegal Move => The selected piece cannot move in that direction or is
blocked by another piece.

Check Violation=> Your King is in Check. You must make a move to save your
King.

Out of Bounds=> You cannot move your piece out of the 10 x 8 board.

Invalid Selection=> You did not select a piece or selected your opponent's
piece.
```

Fig. 3.4: Possible outputs if the move is not legal

3.5 King Check Function

Checks to see if the King is currently in check. If the king is in check, the user/bot is only allowed to move in a way that makes it so the King is NOT in check. If the King is not in check, the game continues normally, or if the King is in checkmate the game ends.

```
Check Violation=> Your King is in Check. You must make a move to save your
King.
```

Fig. 3.5: Possible output if the player tries to make a move leaving the King in Check

3.6 King Checkmate Function

If the King is in checkmate, then the game ends. The user/bot is declared the winner, and the user is returned to the main menu.

```
Game over - Player 1 Wins!

Game over - Player 2 Wins!

Game over - Computer Wins!
```

Fig. 3.6: Possible outputs if King is in checkmate

3.7 Promotion Function

Checks if an ant/pawn has made it to the edge of the board, then it can be promoted to any chess piece, including the anteater. The user will be able type in which piece the ant/pawn is promoted to, and the program will display the replaced piece on the board. The Computer will automatically always choose to upgrade to a queen.

```
Ant has reached the edge of the board - Choose which piece you'd like to
promote it to:

  1. Rook
  2. Knight
  3. Bishop
  4. Anteater
  5. Queen

Please input the number of which one:
```

Fig. 3.7: Output for ant promotion when it reaches the end of the board

3.8 Castling Function

Castling can only occur when the Rook and King have not moved from their initial positions. A check will be made for movement. If neither has moved, and the user inputs that they wish to castle, then castling can occur. If either the Rook or King has moved, this action cannot occur. Additionally, castling cannot occur if the King is currently in check or if the King would pass through or land on a square that is under attack. It is done by choosing the coordinates of both the rook and the knight.

```
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    | wP | wP |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    | wP |    |    | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP |    | wA | wB | wN |    |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK |    |    |    | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J
```

→

```
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    | wP | wP |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    | wP |    |    | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP |    | wA | wB | wN |    |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ |    | wR | wK |    |    |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J
```

```
Type the coordinate of which piece to be moved in format of 'A2'(or 1/2/3):
F1 J1
```

Fig. 3.8: Concept of castling function sequence

3.9 En Passant Function

If an ant/pawn moves two squares forward from its starting position and lands beside an enemy ant/pawn, the opponent has the option to capture it "en passant" on the very next turn. The program checks whether this special capture is available and presents it as a legal move. If the opponent does not take it on that turn, the opportunity is lost.

```
  +----+----+----+----+
8 |    |    |    |    |
  +----+----+----+----+
7 |    |    |    |    |
  +----+----+----+----+
6 |    |    |    |    |
  +----+----+----+----+
5 | bP |    |    |    |
  +----+----+----+----+
4 |    | bP |    |    |
  +----+----+----+----+
3 |    |    |    |    |
  +----+----+----+----+
2 | wP | wP |    |    |
  +----+----+----+----+
1 |    |    |    |    |
  +----+----+----+----+
    A    B    C    D
```

```
  +----+----+----+----+
8 |    |    |    |    |
  +----+----+----+----+
7 |    |    |    |    |
  +----+----+----+----+
6 |    |    |    |    |
  +----+----+----+----+
5 | bP |    |    |    |
  +----+----+----+----+
4 | wP | bP |    |    |
  +----+----+----+----+
3 |    |    |    |    |
  +----+----+----+----+
2 |    | wP |    |    |
  +----+----+----+----+
1 |    |    |    |    |
  +----+----+----+----+
    A    B    C    D
```

```
  +----+----+----+----+
8 |    |    |    |    |
  +----+----+----+----+
7 |    |    |    |    |
  +----+----+----+----+
6 |    |    |    |    |
  +----+----+----+----+
5 | bP |    |    |    |
  +----+----+----+----+
4 |    |    |    |    |
  +----+----+----+----+
3 | bP |    |    |    |
  +----+----+----+----+
2 |    | wP |    |    |
  +----+----+----+----+
1 |    |    |    |    |
  +----+----+----+----+
    A    B    C    D
```

Fig. 3.9: En Passant function board display output

3.10 Undo Function

The user can choose to withdraw their previous move. The board will revert to the state it was in before the move was made. This can be selected from the menu during gameplay.

```
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    | wP |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP |    | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J
```

```
[1 to Pause Game, 2 to Undo Previous move, 3 for Hint]

Type the coordinate of which piece to be moved in format of ‘A2’ (or 1/2/3): 2
```

```
Move undone!

   +----+----+----+----+----+----+----+----+----+----+
 8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
   +----+----+----+----+----+----+----+----+----+----+
 7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
   +----+----+----+----+----+----+----+----+----+----+
 6 |    |    |    |    |    |    |    |    |    |    |
   +----+----+----+----+----+----+----+----+----+----+
 5 |    |    |    |    |    |    |    |    |    |    |
   +----+----+----+----+----+----+----+----+----+----+
 4 |    |    |    |    |    |    |    |    |    |    |
   +----+----+----+----+----+----+----+----+----+----+
 3 |    |    |    |    |    |    |    |    |    |    |
   +----+----+----+----+----+----+----+----+----+----+
 2 | wP | wP | wP | wP | wP | wP | wP | wP | wP | wP |
   +----+----+----+----+----+----+----+----+----+----+
 1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
   +----+----+----+----+----+----+----+----+----+----+
     A    B    C    D    E    F    G    H    I    J

[1 to Pause Game, 2 to Undo Previous move, 3 for Hint]

Type the coordinate of which piece to be moved in format of 'A2' (or 1/2/3):
```

Fig. 3.10: Hypothetical undo function choices

3.11 Save Function

The user can save the current state of the board to a file at any point during the game, this option will then prompt the user for a name of the saved game. A saved game can be loaded from the main menu.

```
Paused Menu:

   1. Resume
   2. Start New Game
   3. Save Game
   4. Quit

Please input the number of which option you'd like to choose: 3

'Save Game' has been chosen, Input a file name to save it under: [name]

'[name]' has been saved!

Paused Menu:
```

```
1. Resume
2. Start New Game
3. Save Game
4. Quit
```

Fig. 3.11: Save Function sequence

3.12 Load Function

Displays past saved files, user inputs which file to load, restoring the board and move history exactly as it was left.

```
Welcome to Anteater Chess!

1. Change Difficulty
2. Change Timer
3. New Game
4. Load Game
5. Quit

Please input the number of which option you'd like to choose: 4

Which game file would you like to load? (game1, game2, game3): game1

game1 loaded!
```

Fig. 3.12: Load function sequence

3.13 Log File Function

The program automatically records all moves made during a game to a human-readable log file. The log can be viewed after the game ends and tracks the full move history in order.

```
White: Moved A2 to A4

Black: Moved B7 to B6

White: . . .

Black . . .
```

Fig. 3.13: Format of how Log File would look

3.14 Hint Function

The user can request a hint during their turn. The program will suggest a legal move for the current player based on the current board state.

```
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP | wP | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J

[1 to Pause Game, 2 to Undo Previous move, 3 for Hint]

Type the coordinate of which piece to be moved in format of ‘A2’ (or 1/2/3): 3

Move a pawn forward!

Type the coordinate of which piece to be moved in format of ‘A2’ (or 1/2/3):
```

Fig. 3.14: Concept of hint sequence

3.15 AI Move Function

When it is the computer's turn, the program calculates all legal moves available, evaluates them, and selects the best one. The computer will make its move within a reasonable time (under 1 minute).

```
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP | wP | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
     A    B    C    D    E    F    G    H    I    J
```

```
Computer's turn.
```

```
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    | wP |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP |    | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
     A    B    C    D    E    F    G    H    I    J
```

```
Computer moved from F2 to F3.
```

Fig. 3.15: Sequence for when it's the computer's turn

3.16 Difficulty Selection Function

The user can choose the difficulty level of the AI opponent - beginner, intermediate, or expert - from the main menu before starting a game. Higher difficulty levels result in smarter and more strategic computer moves.

- Beginner mode AI will not know to do the Castling or En Passant Function.
- Intermediate AI will only know En Passant
- Expert AI will know all possible moves & choose best

```
Welcome to Anteater Chess!
```

```
1. Change Difficulty
2. Change Timer
3. New Game
4. Load Game
5. Quit

Please input the number of which option you'd like to choose: 1.

Choose computer's difficulty option:

1. Beginner
2. Intermediate
3. Expert

Please input the number of which option you'd like to choose: [#]

Computer difficulty updated!
```

Fig. 3.16: Sequence for when user would try to change computer

3.17 Pause Function

The user chooses the pause option during the game, timer pauses; choosing to unpause resumes timer. If the game is paused for 15 minutes, it will close the program.

```
Game Paused, [#] seconds left

Paused Menu:

1. Resume
2. Start New Game
3. Save Game
4. Quit

Please input the number of which option you'd like to choose:
```

Fig. 3.17: Output for when program paused

3.18 Start Function

Displays options of Difficulty Selection, New Game, Timer (1 minute by default), Load Game, or Quit. Difficulty Selection chosen would run the Difficulty Selection Function; New Game would display a reset board; Timer allows the user to choose how long each player has to think of which move they're going to do; Load Game would display the previously saved games and give an option to choose one of them to load; Quit closes the program.

```
Welcome to Anteater Chess!

   1. Change Difficulty
   2. Change Timer
   3. New Game
   4. Load Game
   5. Quit

Please input the number of which option you'd like to choose:
```

Fig. 3.18: Output for when program starts running

Troubleshooting

The following are common issues users may encounter:

- Program does not launch: Ensure you are on a compatible Linux server (crystalcove, zuma, bondi, laguna, balboa, or doheny) and that the binary has execute permissions. Run chmod +x chess and try again.
- Make fails during build: Confirm that gcc is installed and that you are in the correct directory containing the Makefile.
- Save file does not load: Ensure the save file is in the same chess/ directory and that the filename is entered exactly as saved.
- Board does not display correctly: Your terminal window may be too small. Try resizing it or using a different terminal

Error Messages

The following error messages may appear during gameplay. Each describes the cause and the expected program behavior:
Illegal Move => The selected piece cannot move in that direction or is blocked by another piece.
Check Violation=> Your King is in Check. You must make a move to save your King.
Out of Bounds=> You cannot move your piece out of the 10 x 8 board.
Invalid Selection=> You did not select a piece or selected your opponent's piece.
Please select from one of the numbers shown=> Invalid input, input one of the shown numbers from the menu.

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References

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